

Cable hang: comparison of simulation methods

2 supports, $L = 1.000$ m, $S = 0.800$ m

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1 Setup

A cable of length $L = 1.000$ m hangs from 2 supports, all at the same height $z = 1.000$ m, evenly spaced across a horizontal span $S = 0.800$ m. Every method solves this identical scenario, discretised into 60 segments, and each is initialised on the analytic catenary so that what is measured is how far a solver *drifts* from equilibrium rather than how it recovers from an arbitrary transient.

Cable. $r = 2.00$ mm, $E = 80.0$ MPa, $\nu = 0.45$, $\rho = 2390$ kg/m³, mass 30.0 g ($\mu = 30.0$ g/m), so $EI = 1.005e - 03$ N m² and $EA = 1.005e + 03$ N.

2 Analytic reference

For a perfectly flexible, inextensible cable on equal-height supports,

$$z(x) = H + a \left[\cosh\left(\frac{x - S/2}{a}\right) - \cosh\left(\frac{S}{2a}\right) \right], \quad L = 2a \sinh\left(\frac{S}{2a}\right), \quad (1)$$

with $a = T_0/w$ solved from the length constraint. Because all supports share one height, the 2-support case separates exactly into 1 independent sub-catenaries of span $S/1$ and length $L/1$. For this scenario $a = 0.3382$ m and the sag is 265.4 mm.

Validity. The catenary is the $EI \rightarrow 0$ limit. Bending competes with gravity over the elasto-gravitational length $\ell = (EI/w)^{1/3} = 150.5$ mm. Here $\ell/S_{\text{span}} = 0.188$, which is *not* small. The catenary is being used outside its range of validity and the residuals below should be read as the physical difference between an elastic rod and an ideal chain, not as solver error.

3 Results

Method	RMSE [mm]	max err [mm]	sag [mm]	sag err [mm]	arc drift [%]	wall [s]	settled
Newton cable (add_rod + VBD)	1.29	6.87	267.2	1.8	0.215	139.6	–
PhysX capsule chain (D6)	0.69	1.17	264.8	-0.7	0.034	10.5	–
Warp XPBD rod	2.15	4.91	266.4	1.0	0.043	0.6	✓

Table 1: Every method against the analytic catenary. Reference sag 265.4 mm. *Arc drift* is the change in the cable’s own length during the run and is reference-free: it measures violation of inextensibility directly.

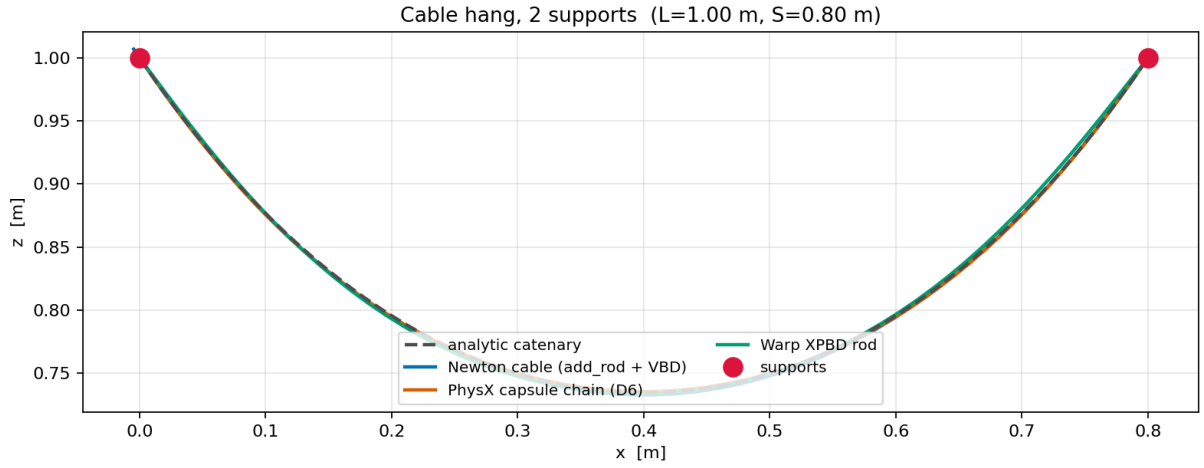


Figure 1: All methods against the analytic catenary.

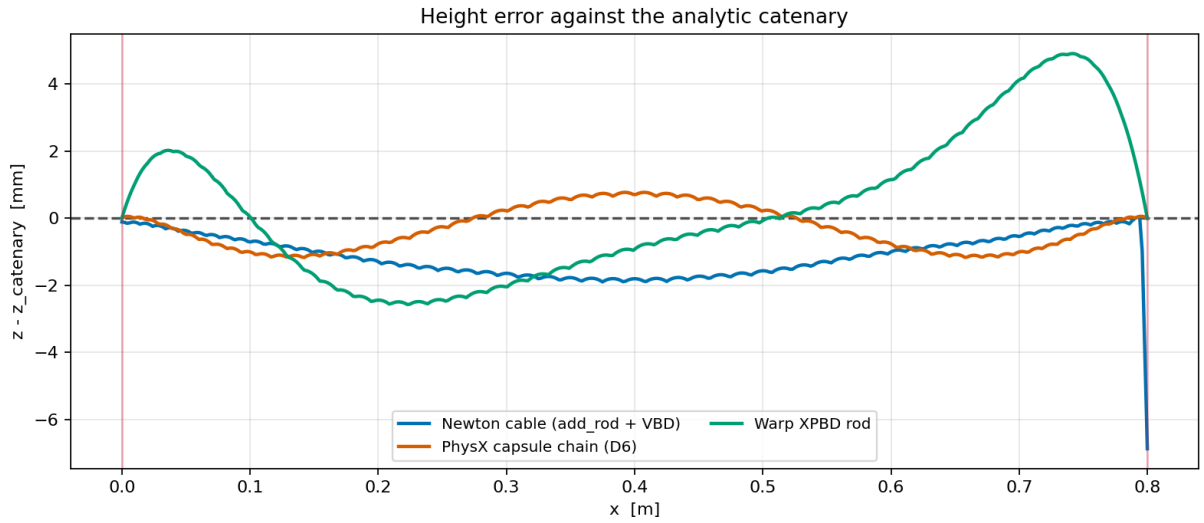


Figure 2: Height error against the analytic catenary. Vertical lines mark the supports.

4 Reading these numbers

- **Arc drift is the honest accuracy check.** It needs no reference shape. A solver that stretches the cable has failed regardless of how catenary-like its profile looks.
- **Agreeing with the catenary is not automatically a win.** Methods with no bending stiffness (a spherical-joint chain, for instance) match the ideal catenary because they share its idealisation, not because they model the cable more faithfully.
- **An unsettled run is provisional.** Where *settled* is blank the motion had not decayed within the time cap, and the profile is a time-average over the final window rather than a true equilibrium.